Deep Learning: Advanced Models and Methods (DL26) • **G23**

Aligning Small LLMs with GRPO for Strict JSON Generation

Rule-Based Reward Optimization for Schema-Conformant Output

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github.com/GiuseppeBellamacina/GRPO-strict-generation

Small LLMs (<2B) Cannot Reliably Produce Valid JSON

- **31–39% Pass@1** for SmolLM2-135M
  (39% No-Think, 31% with Think prompting)
- Unclosed brackets, type violations, conversational filler mixed with output
- **Goal:** use GRPO with rule-based rewards to reach >90% Pass@1 — no neural reward model
- **Dense, deterministic, interpretable** signal drives alignment on a single GPU

Typical failure

```
Sure!  Here's the JSON you asked for:
{   "name":  "Alice",   "age":  30,
"skills":  ["Python", "ML" }
// Missing ] and extra text
```

Expected output

```
“‘json {"name":  "Alice", "age":  30,
"skills":  ["Python", "ML"]} “‘
```

5 Models (135M–2B), Synthetic Dataset, Single GPU

Model	Params	Arch
SmolLM2-135M-Instruct	135M	LLaMA-like
SmolLM2-360M-Instruct	360M	LLaMA-like
Qwen2.5-0.5B-Instruct	0.5B	Qwen2.5
TinyLlama-1.1B-Chat	1.1B	LLaMA 2
Gemma-2-2B-it	2B	Gemma 2

Configuration

- 4-bit NF4 quantization
- LoRA $r=16$, $\alpha=32$
- $1\times$ NVIDIA L40S (48 GB)
- 2 500 steps / model
- Paged AdamW 8-bit
- LR 5×10^{-6}, cosine

Dataset

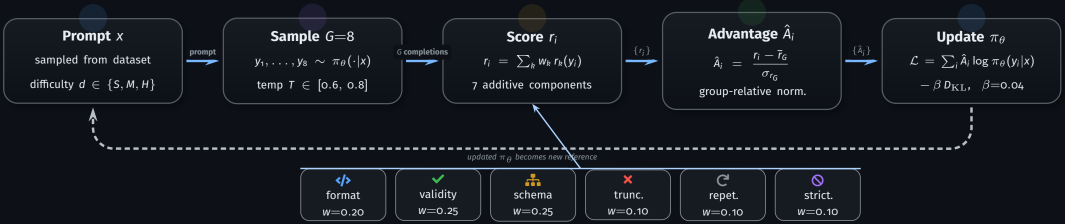
- 24 templates (8/difficulty)
- 4 500 training samples
- 300 eval (100 S / 100 M / 100 H)
- No ground-truth JSON

4 strategies tested:

Think / No-Think $\times$ Standard / Curriculum

Identical hyperparameters across all models

GRPO: Group-Relative Advantage Eliminates the Critic



Key insight | No critic network needed. Advantages are computed *within* each group of $G=8$ completions. Additive reward ensures gradient $\neq 0$ — avoids zero-advantage collapse.

7 Additive Rewards + 3-Stage Curriculum Prevent Collapse

Component	Range	w	Measures
format	[0, 1]	.20	“‘json“‘ fence
validity	[0, 1]	.25	JSON parseability
schema	[0, 1]	.25	Keys, types, nesting
truncation	[−1, 1]	.10	Unclosed braces
repetition	[−1, 1]	.10	Token loops
strictness	[−1, 1]	.10	Extra text outside JSON
reasoning	[−0.5, 1]	.25*	<think> quality

*Think mode only; weight redistributed when off

Curriculum = safety net

Prevents regression in strong models.
Gemma-2-2B: −1.67pp (standard) → +0.67pp (curriculum)

Graduated partial credit

Binary validity gives no gradient for near-valid JSON. Position-based scoring (0.20–0.70) keeps learning alive.

Stage 1: Format (800 st., S:35% M:55% H:10%) › **Stage 2:** Progressive (800 st., S:15% M:35% H:50%) › **Stage 3:** Full (900 st., S:10% M:25% H:65%)

All 5 Models Converge to 87–99% Pass@1 After 2 500 Steps

Model	Pre	Peak	Best Config	Δ
SmolLM2-135M	31%	**90.00%**	Think / Standard	+58.67 pp
SmolLM2-360M	78%	**96.67%**	No-Think / Standard	+19.00 pp
Qwen2.5-0.5B	92%	**98.00%**	Think / Curriculum	+6.33 pp
TinyLlama-1.1B	79%	**99.33%**	No-Think / Curriculum	+20.00 pp
Gemma-2-2B	96%	**97.67%**	Think / Curriculum	+1.33 pp

Best config per model. Pre = same-config baseline.

+58.67 pp

largest gain

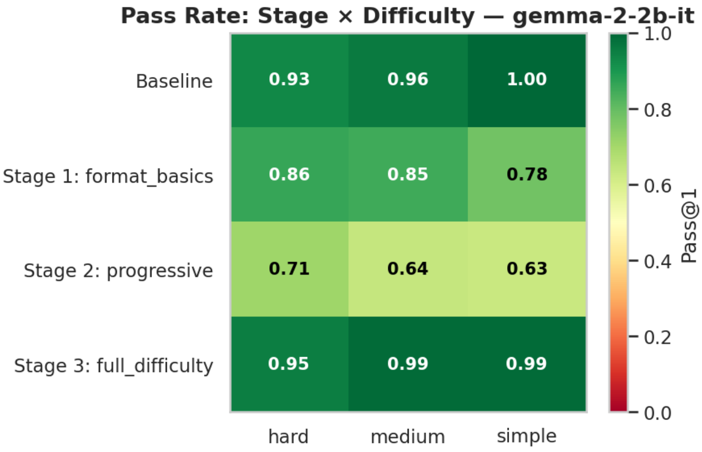
SmolLM2-135M: 31% → 90%

99.33%

best overall

TinyLlama-1.1B (No-Think / Curriculum)

Gemma-2-2B Temporarily Collapses: Curriculum Pacing Must Match Capacity



Gemma-2-2B • Think / Curriculum • Pass@1 per stage & difficulty

U-shaped collapse

96% → 83% → 66% → 98%

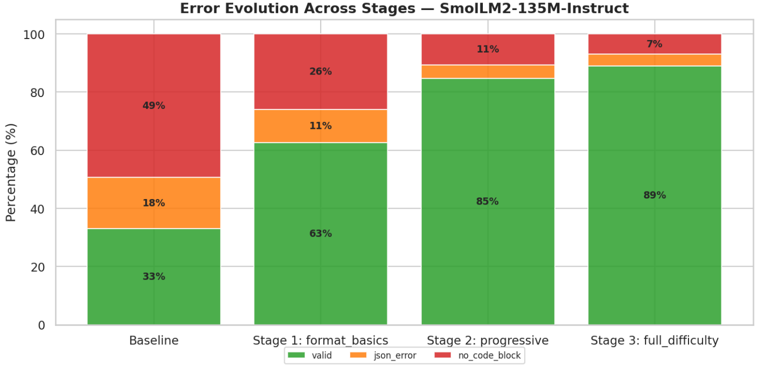
Forcing <think> on trivial schemas confuses an already-strong model.

Full recovery only once hard schemas dominate (Stage 3).

Lesson

Strong models need less warm-up. Standard: −1.67pp; Curriculum: +0.67pp.

Errors Vanish Stage by Stage: Format First, Then Structure



SmolLM2-135M • No-Think / Curriculum • Error types per stage

Stage 1

Dominant error: `no_code_block`

Model learns the “‘`json`“‘ fence first

Stage 2

Format mastered.

Residual: `json_error`

Focus shifts to structure

Stage 3

Near-zero errors.

Only edge-case schema failures remain

Reward Design Matters More Than Model Size

Main Message

GRPO with 7 additive rule-based rewards brings **all 5 models** (135M–2B) to **87–99% Pass@1** on strict JSON generation, single GPU, identical hyperparameters.

Limitations

- JSON-only — not directly generalizable
- Fully synthetic dataset; real-world gap unknown
- Fixed hyperparameters across models

Takeaways

1. **Standard** optimal for weak models (135M: +58.67pp)
2. **Curriculum** safety net for strong (prevents Gemma regression)
3. **Think mode** $<$2pp marginal (small models can't <think>)

Future Work

- Extend to YAML, XML, SQL output
- Real-world API-call datasets
- Reward ablation comparison
- Per-model hyperparameter tuning

Cross-Strategy Insights

Think vs. No-Think

Small models (135M, 360M, 0.5B) never generate <think> tokens; only Gemma and TinyLlama can. 135M peaks 90% in Think by exploiting other rewards, not reasoning.

Curriculum = Safety Net

Produces absolute highest scores (TinyLlama **99.33%**, Qwen **98.00%**). Standard causes Gemma to regress.

Inverted-U Gain Pattern

Smallest models gain most: 135M **+58pp**. Mid-range: +13–26pp. Already-strong: +0–8pp (ceiling effect).

Hard Tier Drives Improvement

Baselines: 17–20% hard pass for 135M. Post-GRPO: **86–99%** on hard prompts. Simple saturates to 100% in Stage 1.

Backup Slides

Additional detail for Q&A

Baseline Pass@1 — Per-Difficulty Breakdown

Model	Overall	Simple	Medium	Hard
SmolLM2-135M	39.00%	58.00%	39.00%	20.00%
SmolLM2-360M	77.67%	85.00%	86.00%	62.00%
Qwen2.5-0.5B	92.67%	95.00%	95.00%	88.00%
TinyLlama-1.1B	79.33%	89.00%	76.00%	73.00%
Gemma-2-2B	**98.33%**	**100%**	**100%**	**95.00%**

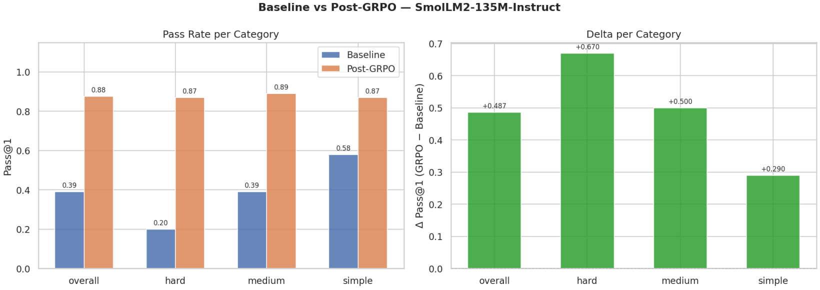
300-sample balanced eval (100 S / 100 M / 100 H). Config: No-Think / Standard.

Reward System — Full Think/No-Think Weight Table

Component	Range	Think	NoThink	What it measures
format	[0, 1]	0.10	0.20	“‘json“‘ fence → 1.0 | generic fence → 0.5 | none → 0.0
validity	[0, 1]	0.15	0.25	JSON parseability; partial credit by error position (0.20–0.70)
schema	[0, 1]	0.20	0.25	Keys, types, nesting depth, required fields
reasoning	[−0.5, 1]	0.25	0.0	<think> block quality; penalises missing/copied
truncation	[−1, 1]	0.10	0.10	Unclosed braces / unterminated strings
repetition	[−1, 1]	0.10	0.10	Token loops, repeated lines, duplicate blocks
strictness	[−1, 1]	0.10	0.10	Extra text outside JSON >50% of output

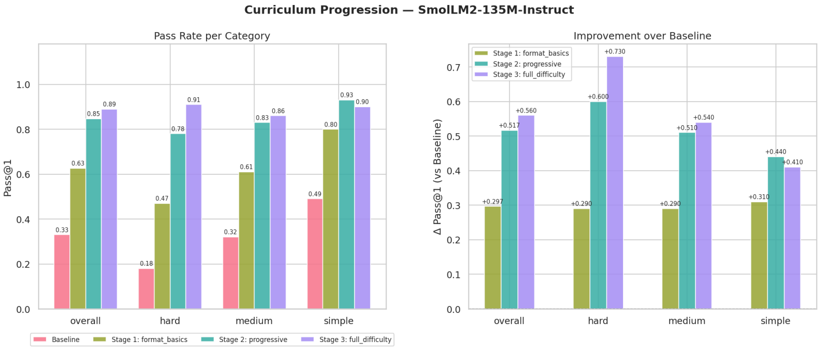
> **Design principle** | Additive composition, no hard gating. Gradient is always non-zero; avoids zero-advantage collapse. When Think is off, reasoning weight is redistributed proportionally.

SmolLM2-135M — No-Think / Standard Comparison



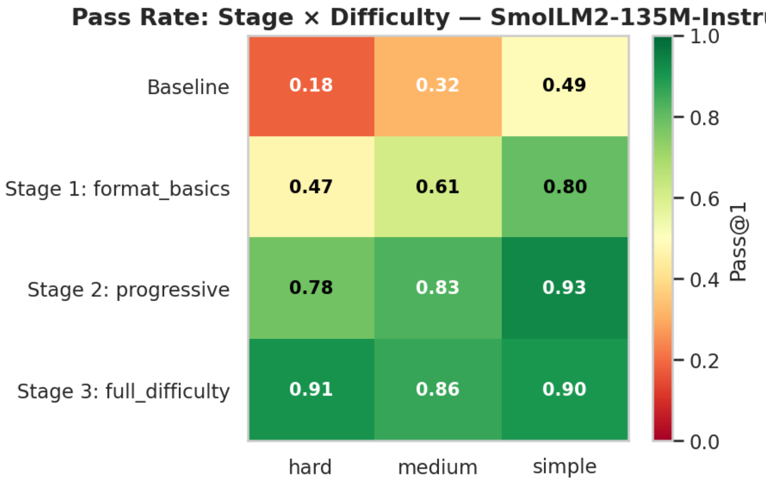
SmolLM2-135M • No-Think / Standard • Pass@1 by difficulty tier, baseline vs. GRPO

SmolLM2-135M — Curriculum Progression (No-Think)



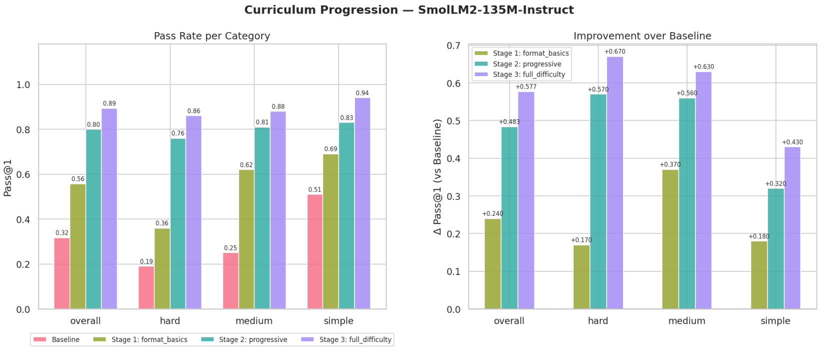
SmolLM2-135M • No-Think / Curriculum • 33% → 62.7% → 84.7% → 89.0%

SmolLM2-135M — Difficulty Heatmap (No-Think / Curriculum)



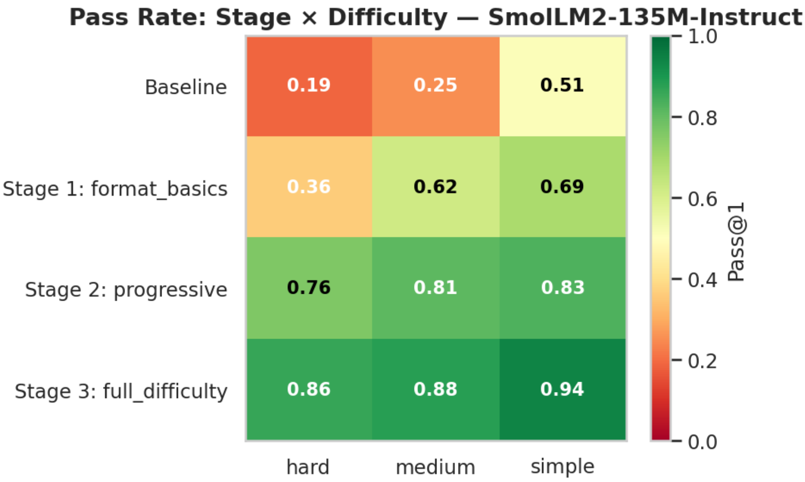
SmolLM2-135M • No-Think / Curriculum • Pass@1 per difficulty tier at each stage

SmolLM2-135M — Curriculum Progression (Think)



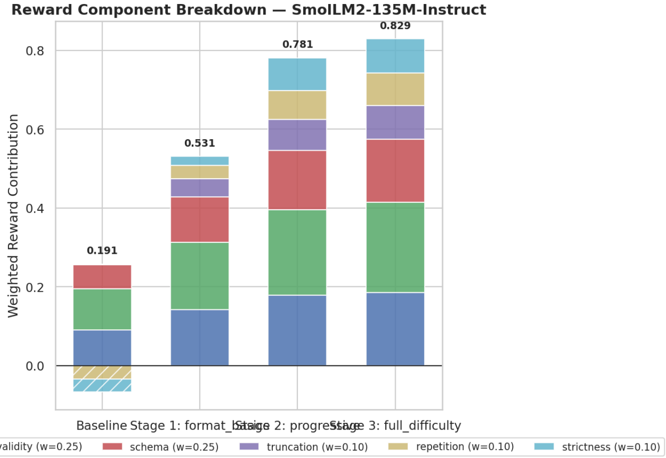
SmolLM2-135M • Think / Curriculum • 31.7% → 55.7% → 80.0% → 89.3%

SmolLM2-135M — Difficulty Heatmap (Think / Curriculum)



SmolLM2-135M • Think / Curriculum • Pass@1 per difficulty tier at each stage

Reward Component Contributions at Final Checkpoint



SmolLM2-135M, No-Think / Curriculum